University of Trento
Master's Degree in Computer Science

Data Visualization Proposal

Data Visualization Lab
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1 Context

1.1 Curiosity Origin

As frequent travelers, we are curious about urban mobility and how people navigate large cities. We became particularly interested in understanding patterns within the context of sustainable transportation and green alternatives, recognizing their growing importance in urban planning and environmental goals. Specifically, we wanted to explore the underlying choices people make when selecting travel modes and the factors that shape those decisions.

This curiosity led us to explore data sources that could provide meaningful insights into urban movement. Among them, New York City's bike rental data stood out as a rich source of information on how people travel across the city. When combined with other datasets, it can offer a clearer picture of the city's daily dynamics.

We also learned that this data is provided by a private company (`Citi Bike`) that operates the bike-sharing service in New York City. For this reason, making our analysis useful and relevant to the company is an effective way to maximize its practical impact.

1.2 Project Circumstance and Purpose

Being one of the largest bike-sharing programs in the world, `Citi Bike` currently collects vast amounts of operational data that, if effectively analyzed and visualized, could provide valuable insights to optimize their services and enhance the user experience.

The primary objective of our visualization tool is to transform this raw operational data into clear, actionable insights for stakeholders.

1.2.1 Citi Bike Operations Team

Specifically, we aim to reveal:

1. **Temporal patterns**: peak usage hours, seasonal trends, and demand cyclicity.
2. **Spatial hotspots**: high-demand stations, commute corridors, and underutilized infrastructure.
3. **User segmentation**: behavioral differences between members and casual riders.
4. **Rideable type preferences**: trends in vehicle usage (classic vs. electric) across different contexts.

While understanding temporal and spatial patterns allows the operations team to optimize station placement and bike rebalancing, analyzing user behavior and ride preferences helps shape targeted marketing and broader service enhancements. Ultimately, focusing on these areas ensures our findings directly support the company's operational efficiency and strategic long-term planning.

1.2.2 City Planners and the General Public

Beyond immediate corporate operations, our tool provides macro-level insights for city planners. By mapping out daily commute flows and transit patterns, the visualization can directly inform municipal infrastructure investments and urban development decisions.

Finally, for the general public, making this data accessible helps raise awareness about urban mobility choices and underscores the tangible benefits of sustainable transport, fostering a more informed and eco-conscious community.

1.3 Intended Message

Our visualization is designed as an **exploratory** analysis tool. Its goal is to help users discover how bicycle-sharing usage is shaped by temporal, spatial, environmental, and behavioral factors, and how these dimensions interact to characterize urban mobility.

Rather than leading users toward a single conclusion, the interface encourages them to formulate and investigate their own hypotheses through interactive filtering and coordinated visualizations.

2 Proposed Datasets

Before selecting the datasets used in this project, we explored several mobility datasets from different cities in order to identify a suitable data source. However, we found that many of these datasets were either incomplete, not regularly updated, or lacked the necessary granularity for our analysis.

After careful consideration, we decided to focus on the `Citi Bike` System Data for New York City, which is well maintained and reflects the complexity of the city.

2.1 Citi Bike System Data

The Citi Bike System Data is published monthly by *Lyft*, the operator of New York City's bike-sharing service. Available for download in CSV format, these open-data trip records date back to 2013.

Each trip record includes key attributes such as start and end timestamps, start and end stations, and user type. These variables allow us to analyze both temporal and spatial patterns in bike usage.

We can also access real-time bike availability and station status via public APIs and integrate it into our dashboard for up-to-date insights.

2.2 Weather Data

Weather conditions significantly influence urban mobility.

We will utilize the Open-Meteo API, which provides historical weather data for any location worldwide, including New York City.

The dataset covers more than 80 years of observations with hourly temporal resolution and a spatial resolution between 1 and 11 km. As an open-source service, *Open-Meteo* offers a comprehensive set of variables, enabling us to better evaluate the impact of environmental conditions on the system.

2.3 New York City Datasets

During our exploration, we also considered various datasets available on the NYC Open Data platform. It offers a wide range of datasets, dashboards, and maps related to the city's transportation system, including bike lanes, traffic data, and other contextual information.

However, to maintain focus on the core dynamics of the bike-sharing system, we rely primarily on the `Citi Bike` System Data, which inherently provides a comprehensive view of local rental patterns. To avoid introducing unnecessary complexity, supplementary datasets from NYC Open Data are excluded unless required for essential context or to support specific insights.

2.4 Datasets Limitations

Another important aspect concerns the quality and consistency of the available data. Historical `Citi Bike` records were collected over multiple years, and they may contain missing values, inconsistencies, or differences in formatting and schema across datasets.

Beyond simple cleaning, extracting new features is essential for uncovering meaningful insights, such as calculating trip durations, distances, or aggregating data.

Then, the integration of weather data also introduces challenges, as it requires aligning granularity in both time and space with the bike trip records.

Finally, the large size of the dataset introduces technical challenges that require efficient data handling strategies to manage the volume of information effectively.

3 Solution

3.1 Vision

We deliver an interactive web-based dashboard as the primary output. This allows all users to explore mobility patterns through a unified, user-friendly interface. Our design approach is to present clear, immediate insights at first glance while also allowing deeper exploration for users who want to examine the details.

The dashboard will automatically refresh on a monthly basis, coinciding with the release of new data from the provider. This ensures that the insights remain accurate and relevant over time, while also enabling historical analysis and comparisons across different periods.

Our framing strategy aims to expose all available features, giving users the freedom to explore in depth and draw their own conclusions. However, we also provide some guidance through the design of the interface and the choice of default views.

3.2 Design Philosophy

Our visualization prioritizes clarity and analytical depth over visual polish. We employ a dual-level approach: high-level dashboards present summary insights (e.g., daily ridership totals, top stations), while detailed views enable granular exploration (e.g., hourly breakdowns, user type comparisons). All design choices remain data-centric, avoiding decorative elements or misleading visual encodings that could obscure underlying patterns.

3.3 Early Design Considerations

While specific chart types and encodings will be refined during development, a few design decisions follow directly from the editorial choices above.

- **Map-centric spatial view**: Spatial data is inherently geographic, so an interactive map is the natural primary choice for our project. A heatmap overlay can show station-level demand (Figures 1), while flow lines can illustrate the most common routes between stations (Figure 2).
- **Faceted temporal charts**: Temporal data will be used mainly to compare behavior across different time periods, so faceted line charts or small multiples will allow us to compare patterns across days of the week, hours of the day, or seasons without overwhelming the viewer with too much information in a single chart (Figure 3).
- **Weather and environmental context**: Incorporating weather data lets us analyze how environmental conditions affect bike usage patterns, using ridge plots such as the one sketched in Figure 4.
- **Footprint and carbon impact**: We will provide a simple metric to quantify the environmental benefits of bike-sharing, such as estimated CO2 emissions saved by replacing car trips with bike trips.
- **User and rideable type breakdowns**: Bar charts or stacked area charts can effectively show the distribution of user types and rideable types, supporting comparisons across different contexts (e.g., time of day, weather conditions).
- **Interactive filtering over static views**: Rather than producing separate dashboards for each dimension, we favour cross-filtering: selecting a time range, station, or user type in one chart updates the others, supporting exploratory analysis without overwhelming the interface.

Regarding focus, the map-centric spatial view serves as the main entry point, offering the most intuitive and immediate reading of the data. Other views will take a supporting role in secondary entries, enriching the analysis without cluttering the interface.

3.4 Constraints

In general, obtaining the most value from the data while respecting project constraints is a key challenge. The project must balance the ambition of performing meaningful analyses with practical limitations such as time availability, data quality, and implementation complexity.

In order to manage both data processing and dashboard development within the project timeline, we want to leverage our programming skills to build a tech stack defined as follows:

- **Backend (Python + Polars)**: efficient data processing, cleaning, and REST API development via FastAPI;
- **Frontend (React + D3.js/Deck.gl)**: custom, interactive visualizations tailored to mobility pattern analysis. Deck.gl will handle the map-centric spatial view, while D3.js will handle temporal and categorical charts.

This modular architecture enables rapid iteration during development while remaining scalable as data volumes grow.

A Early Design Mockups

The following figures collect the early visual references that informed the design choices described in the solution section.

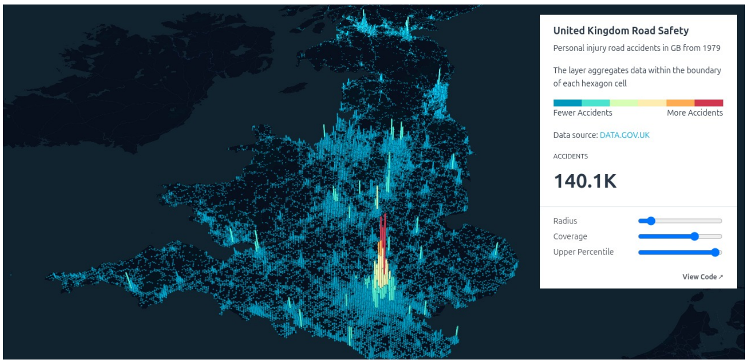

Figure 1: Example of a map-based density layer. In our dashboard, a similar encoding could represent station-level or area-level demand, using height and color intensity to make spatial hotspots immediately visible.

Figure 2: Example of an origin-destination flow layer. For `Citi Bike`, this approach could be adapted to show the strongest station-to-station movements, commute corridors, or directional differences between selected areas.

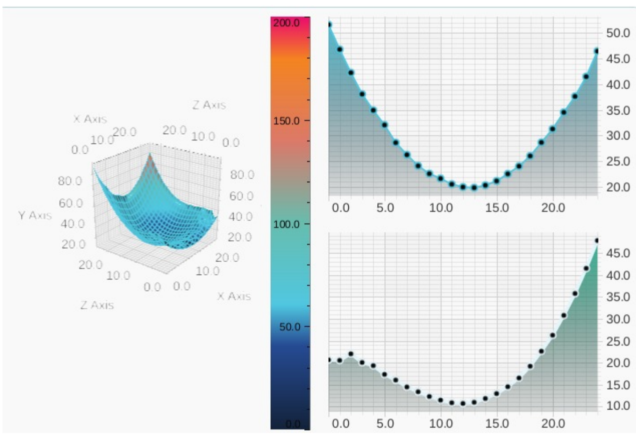

Figure 3: Example of a temporal comparison view. The combination of a surface plot and aligned line charts suggests how we could compare ridership intensity across hours, days, or seasons while still allowing detailed inspection of selected time slices.

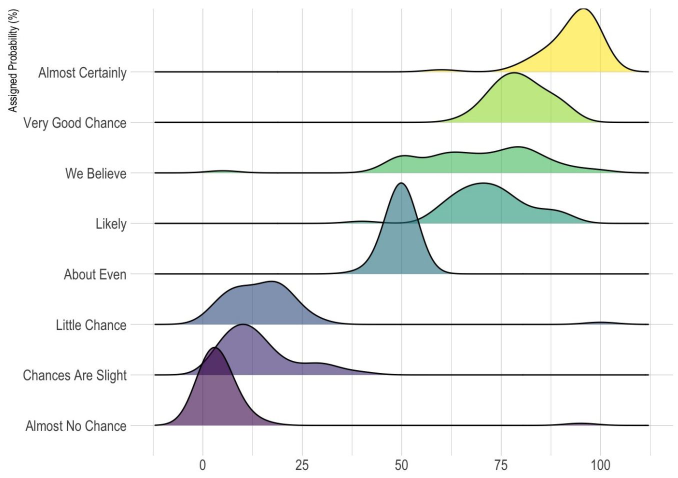

Figure 4: Example of a ridge plot for comparing distributions across ordered categories. In our case, this kind of view could support comparisons of ride volumes or trip durations under different weather conditions, such as temperature ranges, rainfall levels, or wind intensity.